MILITARY SPECIFICATION

FIBER, ACRYLIC, FIBRILLATED

1. SCOPE

1.1 This specification covers one type of Fibrillated acrylic fiber intended for use in 152MM Cartridge cases.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-50250 - Fiber, Acrylic, Fibrillatable

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9235668 - Acrylic Fiber

(copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC: 1320-A-173

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

TAPPI METHOD Technical Association of the Pulp and Paper Industry.
(Application for copies should be addressed to the Secretary, TAPPI, 360 Lexington Avenue, New York 17, N.Y.)

3. REQUIREMENTS

3.1 Material.-The material shall be fibrillated acrylic fiber and shall conform to all applicable drawings and specifications.

3.2 Drainage rate.-The fibrillated fiber shall have a drainage rate of 12.5 ± 2.5 seconds when tested as specified in 4.3.3.1.

3.3 Water content.-The fibrillated fiber shall contain a minimum of 40 percent water when tested in accordance with 4.3.3.3.

3.4 Contamination.-A handsheet formed from the fibrillated fiber shall contain a maximum of 10 sq. mm of foreign matter per square meter when tested in accordance with 4.3.3.4.

3.5 Workmanship.-The fiber shall be produced in a thorough workmanlike manner in accordance with best commercial practice. A handsheet, prepared from the fibrillated fiber, shall be free of knits or nodules.

3.6 Process controls.-The contractor shall, as a minimum, include the following process controls; all data shall be recorded.

3.6.1 Beater water temperature each 1/2 hour of beater operation (see 6.3), consistency of the slurry at the beginning of beating, beater roll to bedplate clearance (see 6.4) and load amperage (see 6.5) each 1/2 hour of beater operation and total beating time.

3.7 First Article Sample.-This specification makes provisions for submittal of a first article sample. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Quantity of material in the lot.
e. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of 40 pounds dry basis of fibrillated acrylic fibers which have been produced by the same production processes which the contractor will use in fulfilling the contract, to a Government facility designated by the contracting officer. A similar sample shall also be submitted whenever a change occurs in the manufacturing process, material used, specification or source of supply, such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-The material shall be inspected for all requirements of this specification.

4.2.3 Rejection.-If the material fails to comply with the applicable requirements, the initial production quantity shall be rejected and the contractor shall take the necessary corrective

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action and submit a new first article sample, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantitites presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall consist of drums of fibrillated acrylic fibers made from one or more batches of fibers that have been subjected to the same processing operations and conditions, by one manufacturer, and in accordance with the same specification and specification revision. A batch shall be that quantity of raw materials which has been subjected to the same unit chemical and physical processes intended to make the final product homogeneous.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part

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of the contractor's inspection system. All examinations shall be conducted at 23 ± 2°C and 50 ± 5% R.H.

4.3.2.1 Fiber

Categories	Defects Inspection	Method of (see 6.2)	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 0.65 percent		
201. Workmanship (see 3.5)		Visual	01001

4.3.2.2 Fibers after packaging (see 5.1)

Categories	Defects Inspection	Method of (see 6.2)	Code No.
Critical: None defined.			
Major: None defined			
Minor:	AQL 1.00 percent		
201. Marking incorrect, misleading, or unidentifiable		Visual	02001
202. Workmanship (see 3.5)		Visual	02002

4.3.3 Testing

4.3.3.1 Drainage Rate (see 3.2), Major defect, Code No. 03001.-A random sample shall be selected from each batch and the drainage rate determined in accordance with TAPPI Method T1002. If the sample fails the applicable requirement, the affected batch shall be rejected.

4.3.3.2 Fiber length, Informational.-Each batch shall be randomly sampled and the fiber length by classification determined in accordance with TAPPI Standard T233 for information. Classification shall be accomplished using screens of 8-, 14-, 48-, and 100- mesh. The percent fiber retained on each screen and the percent fines passing through the 100-mesh screen shall be reported.

4.3.3.3 Water content (see 3.3), Major defect, Code No. 04001.-Each batch shall be randomly sampled after dewatering and just prior to packing. Duplicate samples of 100-grams each (minimum) shall be placed into separate wide-mouth jars (with screw caps). Close the jar immediately to prevent loss of water. Testing shall be as specified in 4.4.1. If either sample fails the applicable requirement, the affected batch shall be rejected (see 6.6).

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4.3.3.4 Contamination (see 3.4), Major defect, Code No. 05001.-Sufficient material shall be randomly selected from the first three batches in each lot to prepare ten (10) handsheets in accordance with TAPPI Standard T205. Each handsheet shall be examined for contamination in accordance with TAPPI Standard T213. If the dirt count exceeds the ammount specified in 3.4 for any handsheet, the affected batch shall be rejected and all remaining batches in the lot shall be tested. Any other batches failing the requirement shall be rejected.

4.4 Test Methods and Procedures

4.4.1 Water Content.-Transfer the wet fiber sample from the closed jar to the weighing pan of a suitably calibrated moisture determination balance. Record the weight of the wet sample to the nearest 0.1 gram. Dry the fiber sample to constant weight using the heat source provided with the balance. Record the dry weight of the sample to the nearest 0.1 gram. Calculate the moisture content of the sample to the nearest 0.5 percent.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-The product shall be packed to afford protection against loss of moisture, damage, and contamination during direct shipment from the supply source to the first receiving activity for immediate use. As a minimum, packing shall consist of product sealed in a plastic bag and overpacked in a fiber drum.

5.2 Marking.-Unless otherwise specified, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the title, number, revision and date of this specification.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Beater water temperature should not exceed 100°F maximum.

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6.4 Beater roll to bedplate clearance should be not less than 0.070 inch.

6.5 Load amperage should not exceed 85 amperes.

6.6 Batches rejected for water content may be reworked by mixing with fresh water, dewatering again and retesting for water content.

Custodian:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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